What an Object Is

CSCI-121 · Elements of Computer Programming II

Week 1, Day 1 · Tuesday, August 25 · SC 333

What you leave with today

- You can name the **identity**, **state**, and **behavior** of a real thing
- You have broken a problem into objects — **on paper**, in pairs
- You have seen both robots: the tank you get this week, the rover you build in December
- You know how this course runs and what is due

> No Java syntax today. Not one line. You already know how to write a loop — what you do not have yet is a way to decide **what should be an object at all**.

Two robots

	What it is	When
The tank	One Java class. It drives, scans, and fires. It fights other people's tanks.	You can have one **by Friday**
The rover	A real robot that drives a real maze, finds a sample, and brings it back. Built by your team.	**December**

> Today we start with the question that makes both of them possible: **what are the pieces?**

How this course runs

- **Tuesday and Thursday** here · **section** on Wednesday
- Work is handed in through **Git**. You will set that up in section this week
- **AI tools** are allowed on most work, banned on some, required on a couple. Every assignment says which
- A short **quiz most weeks** — on paper, on syntax, start of class
- The textbook is **Think Java**. It is free online. **Do not buy a zyBooks code.**

> If the bookstore emailed you an access code — ignore it. You do not need it for this section.

Before tomorrow: your GitHub username

Every repository you receive this semester is named after your GitHub username. If we do not have it, you do not get the assignment.

Submit it here:

`https://tally.so/r/obLAP1`

No GitHub account yet? Make one first — it is free and takes two minutes.

`github.com/signup`

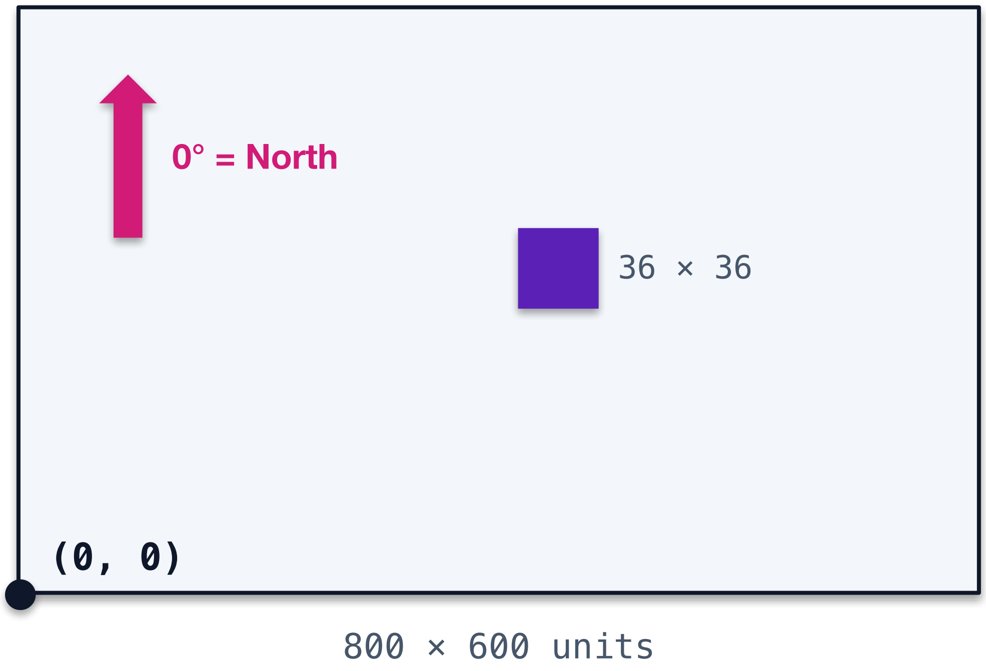
Scan me

> **REQUIRED before tomorrow.** You must have a GitHub account and have submitted your username **before Wednesday's symposium class, which meets online.**

The Arena

The game you get this week — and how it works

The battlefield



- The origin is the **bottom-left** corner — not the top-left, the way most computer graphics work
- **0° is North**, and angles increase **clockwise**. A compass, not your maths class
- Everything is measured in **units**, not pixels
- A tank is **36 × 36**. The arena we fight in is **800 × 600**

> Nobody is driving. Every tank out there is running code somebody wrote **before** the battle started.

A tank is three moving parts

Part	What it does
Body	Drives and turns. Where the tank goes
Gun	Aims and fires — **independently of the body**
Radar	Sweeps for enemies — **independently of both**

The radar reaches **1200 units** and sweeps an arc up to **45°**. The moment the beam crosses an enemy, the tank gets told — *there is someone there, this far away, in this direction.*

Turn the body and the gun and radar get dragged along with it. Steer nothing deliberately and **you stop seeing anything at all.**

> Every scan is already slightly out of date by the time you get it. The enemy kept moving.

Energy is both your health and your ammunition

- Every tank starts a round with **100 energy**
- Firing costs **exactly the power you choose** — from 0.1 up to 3.0 — whether you hit or miss
- A hit **pays you back 3 × the power** you spent
- **0 energy** and you are disabled. **Below 0** and you are gone

Fire at power	It costs you	Damage if it hits	You get back
1.0	1.0	4	3.0
2.0	2.0	10	6.0
3.0	3.0	16	9.0

> Every shot is a bet. Land it and you profit. Miss and you paid for nothing — and a tank that fires at everything runs itself out of energy.

Identity, State, Behavior

Every object, all semester, is these three things

This marker is an object

	The marker	The idea
Identity	It is *this* marker	Twelve identical ones in the box. This is still a specific one.
State	Colour · how much ink · cap on or off	Facts about it right now. **They change.**
Behavior	Writes · dries out · gets capped	What it *does*, or what it is **responsible for**.

Two identical markers

Same colour. Same ink. Same cap. **Same state.**

Are they the same object?

- If you borrowed mine, you owe me **this one** back → **identity** matters
- If you just need a black marker, any will do → only **state** matters

Both answers are right, in different situations. Java has two different tools for those two situations, and you will meet both in September.

> Today, just notice that the question exists.

You are an object

What is your state right now?

- Hungry. Cold. Enrolled in four courses. Sitting in row three.

What is your behavior?

- Not "I walk." Closer to: *I am responsible for getting myself to class.*

Behavior is what a thing is **responsible for** — not just what it can do.

Now the tank on screen

	The tank
Identity	Its name in the battle list. There are three out there — this is one of them.
State	Energy · position · heading · gun heat
Behavior	Drives · scans · fires · reacts to being hit

One person wrote that as **one class**. On Friday you get the same class and change it.

Decomposition

Deciding what should be an object — and what should not

A vending machine — what objects are in here?

Say everything. We will write down the bad suggestions too — the list is supposed to be messy at first.

- Is **price** an object, or state belonging to a snack?
- Is **customer** inside our problem, or outside it?
- Is **slot** a real thing, or just a number?

> Half of decomposition is deciding what is **NOT** an object. A thing with no behavior of its own is usually **state on something else**.

Where we land

Four objects we can defend — and for each one, all three columns:

Object	Why it earns the name
`VendingMachine`	Takes money, dispenses, knows if it is out of service
`Snack`	Has a name and a price; knows nothing about money
`Inventory`	Knows what is left, and where
`CoinBox`	Takes coins, makes change, can be empty

Your turn — the campus laundry room

Students swipe a card to start a washer or dryer. Machines can be **free**, **running**, or **broken**. A load takes a set time. Students want to know from their room whether anything is free.

Find the objects. For each one, list identity, state, and behavior.

- Pairs · paper only · no laptops
- There is **no single right answer**. You are graded on whether you can defend it

> If a thing has no behavior of its own — it is probably state on something else.

Your first class diagram

Washer
status minutesLeft
start() isAvailable()

← **name**

← **state**

← **behavior**

Three compartments:

- the **name**
- its **state**
- its **behavior**

That is a class diagram. Thursday we connect them with arrows. In section you learn to write them as text that GitHub draws for you.

Exit ticket — hand in on your way out

Name one object from your own life outside this room.

Give its **identity**, its **state**, and **one behavior**.

Bottom of your handout. Hand it to me as you leave.

Before you go

- **Wednesday's symposium is online** — and you need a **GitHub account** and your username submitted before it
- Section this week is **Recitation 0** — bring a laptop, we install everything
- **Do not buy a zyBooks code.** The textbook is Think Java, free online
- **AI is allowed** on Skill Builders, with a note saying what you asked for. Twice this term you will sit with me for eight minutes, I will put one bug in code you handed in, and you will fix it in front of me
- **A tank is waiting** if you want one — optional, runs all semester, no hardware

> **Thursday:** how to decide when one object should be two.

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